

A
Realtime Research Project Report
ON
COMPREHENSIVE RETAIL BUSINESS INTELLIGENCE &
SALES PERFORMANCE ANALYSIS AND VISUALIZATION OF
DATA

Submitted in partial fulfilment of Requirements for the Award of the
Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)
SIDDHARTHA INSTITUTE OF TECHNOLOGY & SCIENCES

(UGC-AUTONOMOUS)

(Approved by AICTE, New Delhi & Affiliated to JNTUH, Hyderabad)
Accredited by NBA and NAAC with 'A+' Grade
Narapally, Korremula Road, Ghatkesar, Medchal-Malkajgiri (Dist)-500 088
2024 - 2028

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CERTIFICATE

This is to certify that the project work entitled “**COMPREHENSIVE RETAIL BUSINESS INTELLIGENCE & SALES PERFORMANCE ANALYSIS AND VISUALIZATION OF DATA**” is a bonafide record of project work done jointly by **A ABHISHEK (24TQ1A6701), S SAINATH (24TQ1A6741), M LINGASWAMY (24TQ1A6728), J VIGNESH (24TQ1A6722), V NITHIN (25TQ5A6702)** under my supervision and is submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in **COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)** by Jawaharlal Nehru Technological University Hyderabad (JNTUH), during the academic year 2025-2026

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We declare that this project report titled **CREDIT CARD TRANSACTION ANALYSIS AND FRAUD ANALYSIS USING MACHINE LEARNING** submitted in partial fulfilment of the degree of **B. Tech in CSE(DATA SCIENCE)** is a record of original work carried out by us under the supervision of **Mr P. SRINU** and has not formed the basis for the award of any other degree or diploma, in this or any other Institute or University. In keeping with the ethical practice in reporting scientific information, due acknowledgments have been made wherever the findings of others have been cited.

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Vision of the Department: : To Empower Students With World-Class Knowledge And Technical Expertise In Computer Science And Engineering (Data Science), Fostering Innvoators And Ethical Professionals Who Drive Data-Driven Transformation For a Smater ,Sustainable,And Inculsive Society

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MISSION	STATEMENT
DM1	Deliver a dynamic curriculum grounded in the latest advancements in datascience computational mathematics, and computer science
DM2	Cultivate analytical thinking critical problem solving, and Lifelong Learning through experiential and industry-oriented Education.
DM3	Foster a Culture of innovation research, and enterprenurship That addresses real-world challengesusing data-drivensolutions

Programme Educational Objectives (PEO)

PEO1: Graduates will be able to synthesize mathematics, science, engineering fundamentals, laboratory and work – based experiences to formulate and to solve problems proficiently in Computer science and Engineering and related domains.

PEO2: Graduates will be prepared to communicate effectively and work in multidisciplinary engineering projects following the ethics in their profession.

PEO3: Graduates will recognize the importance of and acquire the skill of independent learning to shine as experts in the field with a sound knowledge.

ABSTRACT

Retail businesses generate vast amounts of data from transactions, customer interactions, inventory, and supply chains. However, extracting meaningful insights from this data remains a significant challenge. This project presents a comprehensive Retail Business Intelligence (BI) system designed to analyze sales performance, optimize operations, and effectively visualize data.

The system leverages data analytics, machine learning, and advanced visualization tools to provide actionable insights, including sales trends, customer behavior, product performance, and revenue forecasting. Interactive dashboards and reports enable decision-makers to monitor key performance indicators (KPIs) in real time.

By integrating data from multiple sources and applying advanced analytical techniques, the platform enhances operational efficiency, increases profitability, and supports data-driven decision-making in retail businesses.

Furthermore, the system incorporates data preprocessing techniques to ensure data quality and consistency before analysis. Advanced algorithms are applied to detect patterns, trends, and anomalies within large datasets, enabling businesses to respond proactively to market changes.

The visualization component plays a crucial role by transforming complex data into intuitive graphs, charts, and dashboards, making it easier for stakeholders to interpret information quickly and accurately. These visual tools support strategic planning, performance monitoring, and operational improvements.

Additionally, the platform supports scalability and flexibility, allowing integration with various data sources such as point-of-sale systems, e-commerce platforms, and supply chain databases. This ensures that the system can adapt to different retail environments and business needs.

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CHAPTER 1

INTRODUCTION

INTRODUCTION

In today's highly competitive retail environment, businesses generate enormous volumes of data from daily operations such as sales transactions, customer interactions, inventory management, and supply chain activities. While this data holds valuable insights, many organizations struggle to effectively analyze and utilize it for strategic decision-making. Traditional methods of data analysis are often time-consuming, fragmented, and lack the capability to provide real-time insights, which limits a retailer's ability to respond quickly to market demands and customer preferences.

Retail Business Intelligence (BI) systems have emerged as a powerful solution to address these challenges by transforming raw data into meaningful and actionable information. This project focuses on developing a comprehensive Retail Business Intelligence and Sales Performance Analysis system that integrates data from multiple sources and applies advanced analytical techniques. The system is designed to evaluate key aspects such as sales trends, customer behavior, product performance, and revenue patterns, enabling businesses to gain a deeper understanding of their operations.

The platform utilizes data analytics, machine learning, and data visualization tools to deliver insights through interactive dashboards and reports. These visual representations simplify complex datasets, allowing decisionmakers to monitor key performance indicators (KPIs) in real time and make informed decisions. Additionally, the system supports optimization strategies by identifying inefficiencies in inventory management, sales processes, and customer engagement.

By providing a unified and intelligent view of retail data, this project aims to enhance operational efficiency, improve profitability, and support data-driven decision-making. Ultimately, the implementation of such a system empowers retailers to stay competitive, adapt to changing market conditions, and deliver better value to customers.

1.1 PROBLEM DEFINITION

Retail organizations often face significant challenges in managing and utilizing their data effectively. One of the major issues is the fragmentation of data across multiple systems, making it difficult to obtain a unified view of business operations. Additionally, the lack of real-time insights prevents timely decision-making, which is crucial in a fast-paced retail environment. Inefficient inventory management further leads to problems such as overstocking or stock shortages, directly impacting sales and customer satisfaction. Moreover, many businesses have a poor understanding of customer behavior, limiting their ability to offer personalized experiences. Traditional reporting systems are often inadequate, as they fail to provide actionable insights, ultimately resulting in revenue loss and ineffective decision-making.

1.2 OBJECTIVE OF THE PROJECT

The primary objective of this project is to develop a comprehensive Business Intelligence system tailored for retail analytics. The system aims to analyze sales performance and identify trends that can help businesses make informed decisions. It also focuses on optimizing inventory and supply chain operations to improve efficiency and reduce costs. Another key objective is to present data through interactive dashboards and reports, making complex information easy to understand. Ultimately, the project seeks to support data-driven decision-making by providing accurate, timely, and meaningful insights to retail businesses.

1.3 SCOPE OF THE PROJECT

The project includes:

- **Data collection from retail datasets (sales, customers, products)**

Collection of structured and unstructured data from multiple retail sources such as POS systems, online platforms, and databases.

- **Data preprocessing and cleaning**

The collected data is cleaned by handling missing values, removing duplicates, and correcting inconsistencies. This step improves data quality and prepares it for accurate and reliable analysis.

- **Sales and customer analysis**

Sales data is analyzed to identify trends, best-selling products, and revenue patterns. Customer behavior is studied to understand preferences, segmentation, and purchasing habits.

- **Dashboard creation for visualization**

Interactive dashboards are developed using visualization tools to represent data through charts and graphs. These dashboards help users easily interpret data and monitor key performance indicators in real time.

- **Predictive analytics for future sales trends**

Machine learning and statistical techniques are used to forecast future sales based on historical data.

1.4 MOTIVATION

Retail businesses are increasingly relying on data to stay competitive in a rapidly evolving market. However, without proper tools and technologies, a large portion of this data remains underutilized, limiting its potential to drive business growth. This project aims to bridge that gap by transforming raw and scattered data into meaningful insights, enabling smarter business decisions and improving overall customer satisfaction.

In today's retail environment, customer expectations are continuously rising, and businesses must respond quickly to changing trends and preferences. Without an effective system to analyze data, retailers often rely on intuition rather than evidence-based strategies, which can lead to inefficiencies, inaccurate forecasting, and missed opportunities. By implementing a Business Intelligence system, this project empowers organizations to gain a clear and detailed understanding of sales performance, customer behavior, and operational efficiency.

Another key motivation for this project is the need to handle the growing complexity and volume of retail data. With the expansion of online shopping, digital payments, and multi-channel sales platforms, data is being generated at an unprecedented rate. Managing and analyzing such large datasets manually is not only timeconsuming but also prone to errors. Therefore, an automated system that can process, analyze, and present data efficiently becomes essential for modern retail businesses.

Furthermore, the project emphasizes the importance of data visualization in simplifying complex information. Through interactive dashboards and graphical representations, stakeholders can easily interpret trends, compare performance metrics, and identify areas that require improvement. This reduces dependency on technical expertise and allows even non-technical users to make informed decisions quickly.

The motivation also extends to improving resource utilization and operational efficiency. By identifying patterns such as slow-moving inventory, peak sales periods, and customer preferences, businesses can optimize stock levels, reduce waste, and enhance supply chain management. This leads to cost reduction and improved profitability.

Additionally, this project is driven by the need to promote a data-driven culture within organizations. Encouraging the use of analytics in everyday decision-making helps businesses become more proactive rather than reactive. It enables them to anticipate market changes, personalize customer experiences, and implement effective marketing strategies.

CHAPTER 2

LITERATURE SURVEY

2: LITERATURE SURVEY

1. Retail Analytics Using Big Data

Retail Analytics using Big Data plays a crucial role in modern retail systems by enabling the analysis of large and complex datasets generated from various sources such as transactions, customer interactions, and supply chains. This approach helps in identifying hidden patterns, customer trends, and market behaviors, thereby improving decision-making and strategic planning for businesses.

With the rapid growth of digital platforms and e-commerce, the volume of retail data has increased significantly, making traditional analysis methods insufficient. Big Data technologies such as distributed storage and parallel processing allow businesses to handle vast datasets efficiently. By leveraging these technologies, retailers can gain deeper insights into customer preferences, optimize pricing strategies, and improve demand forecasting, ultimately enhancing competitiveness in the market.

2. Business Intelligence Systems

Business Intelligence (BI) Systems focus on transforming raw data into meaningful insights through data processing, reporting, and visualization tools. These systems enhance organizational performance by providing accurate and timely information, enabling businesses to monitor key metrics, evaluate performance, and make informed decisions effectively.

Modern BI systems integrate data from multiple sources into a centralized platform, allowing organizations to maintain consistency and reliability in their analysis. They also support real-time reporting and automated data updates, reducing manual effort and errors. By using BI tools, businesses can track key performance indicators (KPIs), identify inefficiencies, and implement strategies to improve productivity and profitability.

3. Sales Forecasting Using Machine Learning

Sales Forecasting using Machine Learning is an advanced approach that utilizes predictive models and algorithms to analyze historical sales data and predict future trends. This helps retailers in demand planning, inventory management, and resource allocation, reducing uncertainties and improving overall efficiency.

Machine learning models such as regression analysis, time series forecasting, and neural networks can identify complex patterns that are not easily detectable through traditional methods. These models continuously learn from new data, improving prediction accuracy over time. As a result, businesses can better prepare for seasonal demand, avoid stock shortages or excess inventory, and make proactive decisions to maximize revenue.

4. Customer Segmentation Techniques

Customer Segmentation Techniques involve categorizing customers into different groups based on their purchasing behavior, preferences, and demographics. This enables businesses to target specific customer groups with personalized marketing strategies, improving customer satisfaction and increasing sales.

Advanced segmentation methods use clustering algorithms and data mining techniques to create more precise customer groups. These insights allow businesses to design customized promotions, recommend relevant products, and enhance customer engagement. Effective segmentation not only improves marketing efficiency but also helps in building long-term customer relationships and brand loyalty.

5. Data Visualization Tools in Retail

Data Visualization Tools in Retail emphasize the use of dashboards, charts, and graphical representations to present complex data in an easy-to-understand format. These tools help stakeholders quickly interpret data, identify trends, and make data-driven decisions, enhancing overall business performance.

Modern visualization tools such as Power BI and Tableau provide interactive features that allow users to explore data dynamically. They support real-time updates, drill-down analysis, and customizable dashboards, making it easier to gain insights from large datasets. By converting raw data into visual formats, these tools improve communication, support faster decision-making, and make data accessible to both technical and non-technical users.

CHAPTER-3

EXISTING SYSTEM

CHAPTER 3: EXISTING SYSTEM

Existing Methods

- **Manual data analysis using spreadsheets**

Retail businesses often rely on spreadsheets like Excel to store and analyze data manually. This process involves entering, filtering, and calculating data using formulas and functions. However, it becomes highly time-consuming and error-prone as the volume of data increases. Managing large datasets in spreadsheets is inefficient and limits advanced analytical capabilities.

- **Basic reporting tools**

Basic reporting tools are used to generate simple reports such as daily sales summaries and transaction records.

These tools provide limited insights and mainly focus on historical data rather than predictive analysis. They lack interactivity and advanced features needed for deep data exploration. As a result, decision-makers are unable to gain comprehensive insights for strategic planning.

- **Traditional ERP systems**

Traditional ERP systems integrate core business processes such as inventory, sales, and finance into a single platform.

They help in maintaining data consistency and improving operational efficiency.

However, most ERP systems are not designed for advanced analytics or real-time data processing. They often lack modern visualization tools and require additional systems for detailed analysis.

3.1 DRAWBACKS

- **Time-consuming analysis**

Analyzing data using traditional methods requires significant manual effort and time. Employees must gather, clean, and process data before extracting any meaningful insights. This slows down decision-making and reduces responsiveness to market changes. As a result, businesses may miss important opportunities due to delayed analysis.

- **Lack of real-time insights**

Existing systems generally provide insights based on historical data rather than real-time updates. This prevents businesses from tracking current performance and reacting quickly to trends. In a fast-paced retail environment, delayed insights can lead to poor decisions. Real-time data access is essential for improving operational efficiency and competitiveness.

- **Limited visualization capabilities**

Traditional tools offer only basic charts and reports with minimal interactivity.

They fail to represent complex data in a clear and understandable format.

This makes it difficult for users to identify patterns, trends, and anomalies effectively.

Advanced visualization is necessary to support better interpretation and communication of data.

- **Poor scalability**

As the volume of retail data grows, traditional systems struggle to handle large datasets efficiently.

They often experience performance issues and slower processing speeds.

This limits their ability to support expanding business operations and data complexity.

Scalable systems are required to manage increasing data and ensure long-term growth.

CHAPTER 4

PROPOSED SYSTEM

CHAPTER 4: PROPOSED SYSTEM

The proposed system is a comprehensive Retail Business Intelligence (BI) platform designed to overcome the limitations of traditional retail data management systems. It integrates data from multiple sources, processes it using advanced analytical techniques, and presents meaningful insights through interactive dashboards. The system aims to provide real-time analysis of sales performance, customer behavior, and inventory status, enabling businesses to make informed and timely decisions. By leveraging data analytics, machine learning, and visualization tools, the platform enhances operational efficiency, reduces manual effort, and supports a data-driven approach to business growth.

Features

• Data Integration

The system collects data from multiple sources such as point-of-sale systems, e-commerce platforms, and databases.

This ensures a unified and consistent dataset for accurate analysis and reporting.

• Sales Analysis

The platform analyzes sales data to track revenue, profit margins, and growth trends over time. It helps identify top-performing products and seasonal variations in sales.

• Customer Analytics

Customer data is analyzed to understand buying patterns, preferences, and segmentation. This enables personalized marketing strategies and improves customer engagement.

• Inventory Optimization

The system monitors inventory levels to prevent stock shortages and overstocking issues. It helps in maintaining optimal stock levels and improving supply chain efficiency.

• Interactive Dashboards

User-friendly dashboards display real-time insights using charts, graphs, and KPIs. These visualizations make it easier for decision-makers to interpret data quickly.

- **Predictive Analytics**

Machine learning models are used to forecast future sales and demand trends. This helps businesses plan resources, manage inventory, and reduce risks effectively.

Benefits

- **Improved decision-making**

Provides accurate and timely insights that help managers make informed business decisions. It reduces reliance on guesswork and improves strategic planning.

- **Increased profitability**

Identifies profitable products, customer segments, and sales opportunities. This helps in maximizing revenue and minimizing unnecessary costs.

- **Better customer understanding**

Analyzes customer behavior to provide deeper insights into preferences and needs. This improves customer satisfaction and retention.

- **Efficient inventory management**

Ensures proper stock levels by predicting demand and reducing wastage. This leads to better resource utilization and operational efficiency.

CHAPTER 5 HARDWAREAND SOFTWARE REQUIREMENTS

CHAPTER 5: SYSTEM REQUIREMENTS

The successful implementation of the proposed Retail Business Intelligence system depends on appropriate hardware and software resources.

5.1 HARDWARE REQUIREMENTS

• Processor

The processor is a critical component responsible for executing data processing and analytical tasks. A minimum of an Intel i5 or equivalent processor is recommended to handle data computations efficiently.

• RAM

Random Access Memory (RAM) plays a vital role in handling data during analysis and visualization. A minimum of 8 GB RAM is required to run tools like Python, Power BI, and database systems smoothly.

• Storage

Storage is required to store datasets, software tools, and generated reports or dashboards. A minimum of 512 GB storage is recommended to accommodate data files and system requirements.

5.2 SOFTWARE REQUIREMENTS

• Python

Python is the primary programming language used for data analysis, preprocessing, and machine learning. It provides a simple and efficient coding environment with a wide range of libraries.

• Power BI

Power BI is a powerful data visualization tool used to create interactive dashboards and reports. It helps in converting complex data into meaningful visual insights.

- **SQL (Structured Query Language)**

SQL is used for managing and querying structured data stored in databases. It allows efficient data retrieval, filtering, and manipulation.

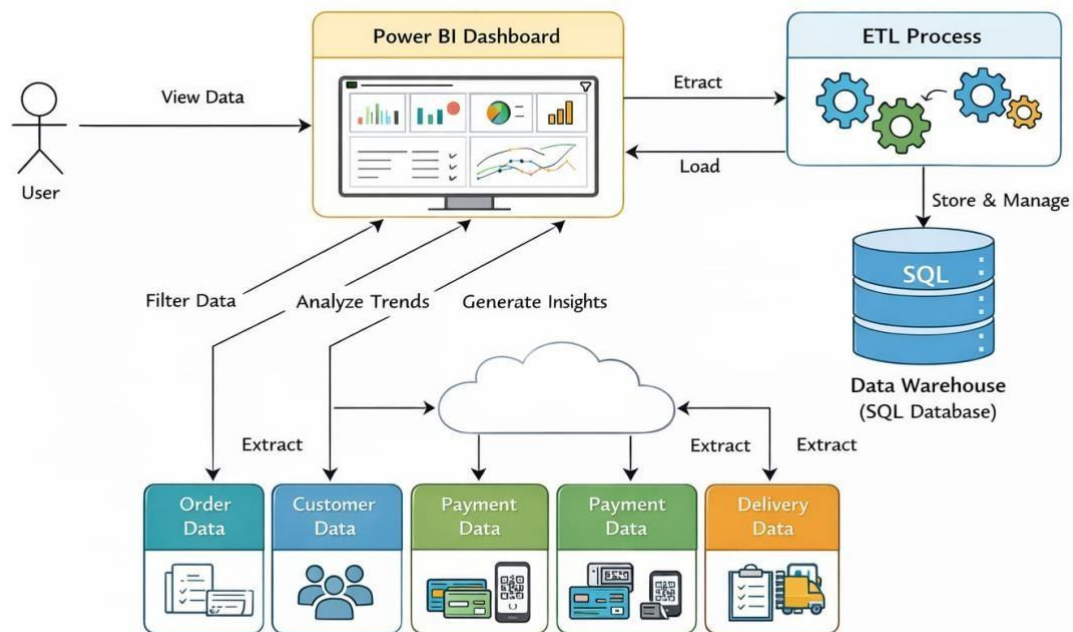
CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM DESIGN

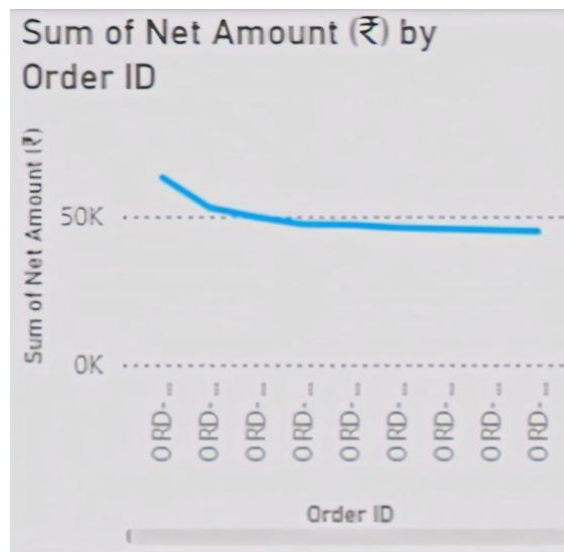
System design defines the structure and workflow of the proposed Retail Business Intelligence system. It outlines how data is collected from multiple sources such as sales, customers, and inventory, and then processed into meaningful information. The system follows a layered approach, including data collection, preprocessing, storage, analysis, and visualization. Data is cleaned and transformed using analytical tools, stored in databases, and analyzed using statistical and machine learning techniques. Finally, the processed data is presented through interactive dashboards and visualizations using tools like Power BI. This design ensures efficient data handling, real-time insights, and supports effective decision-making in retail operations.

System Design of Retail Sales Analytics Solution



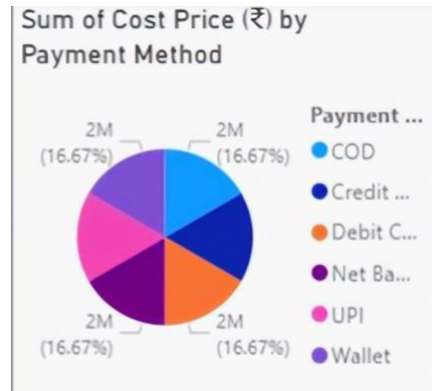
6.1.1 Line Chart

The line chart is used to represent sales trends over a period of time. It helps in identifying patterns such as seasonal variations, growth trends, and fluctuations in revenue. By analyzing the line chart, businesses can understand how sales change over time and make strategic decisions accordingly.



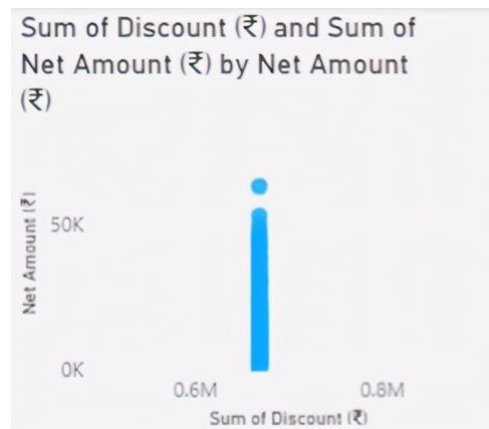
6.1.2 Pie Chart

The pie chart is used to show the distribution of sales across different categories such as products, regions, or customer segments. It provides a clear visual representation of proportional data. This helps in identifying which category contributes the most to overall sales and where improvements are needed.



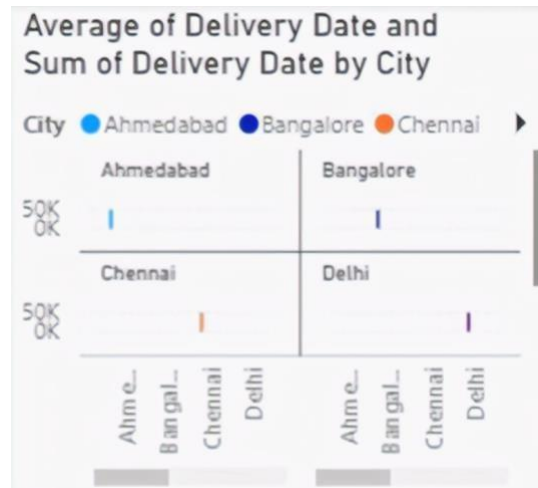
6.1.3 Line and Stacked Column Chart

This chart combines both line and bar representations to show multiple data trends in a single view. It is useful for comparing sales performance along with other factors such as profit or quantity sold. It provides a comprehensive understanding of relationships between different variables.



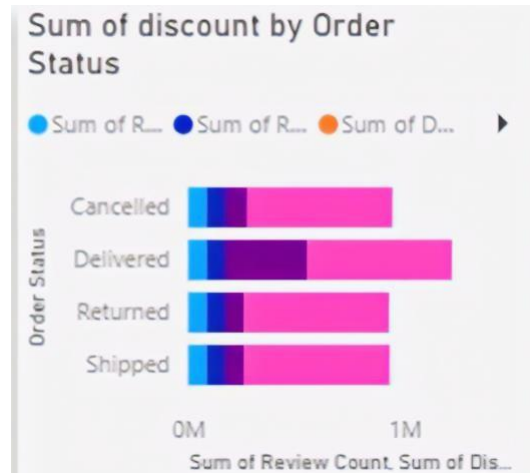
6.1.4 Clustered Column Chart

The clustered column chart is used to compare values across different categories. It is useful for analyzing sales performance of various products or regions side by side. This helps in identifying top-performing and underperforming areas.



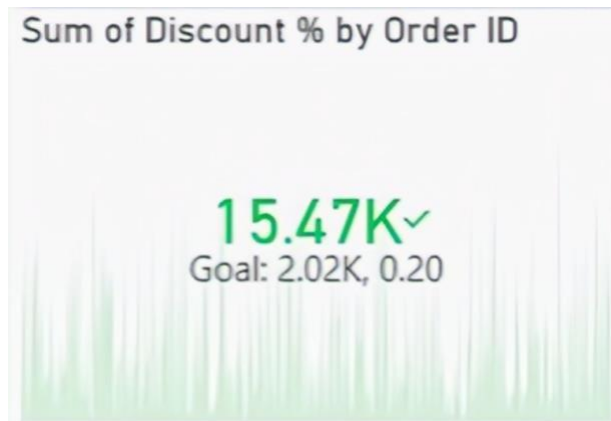
6.1.5 Clustered Bar Chart

The clustered bar chart presents data horizontally, making it easier to compare large sets of categories. It is commonly used for comparing customer segments or product performance. It improves readability when dealing with multiple data points.



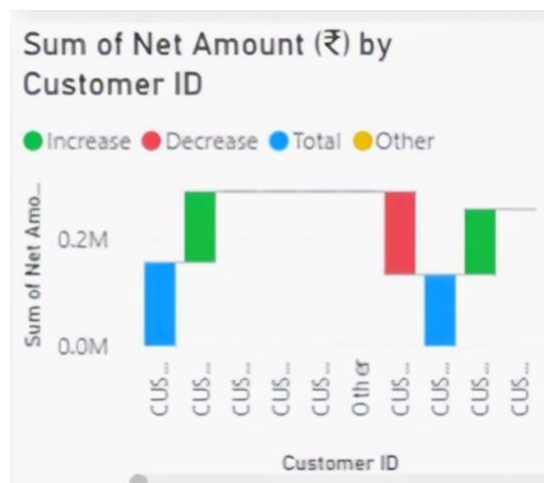
6.1.6 KPI (Key Performance Indicator)

KPI visuals are used to display important business metrics such as total sales, profit, and growth rate. These indicators provide a quick summary of overall performance. They help decisionmakers monitor business health and take immediate action if needed.



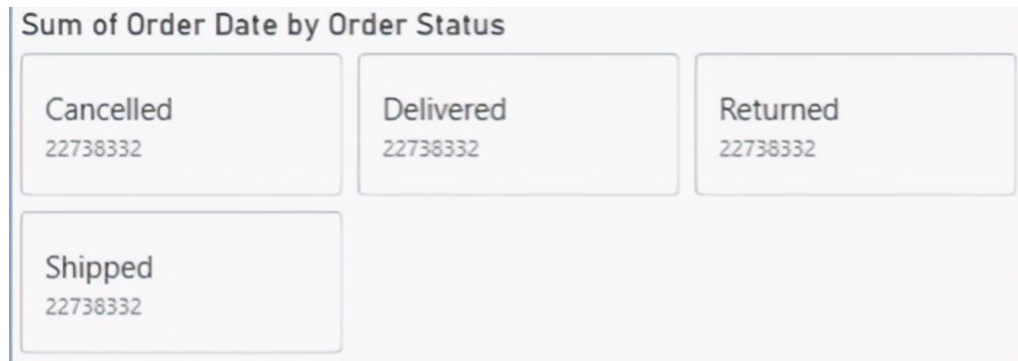
6.1.7 Waterfall Chart

The waterfall chart shows the cumulative effect of sequential data, such as profit and loss over time. It helps in understanding how different factors contribute to the final outcome. This visualization is useful for analyzing revenue changes and identifying key contributors.



6.1.8 Slicers (Button & List Slicer)

Slicers are interactive filters that allow users to dynamically explore data. Button slicers and list slicers enable users to filter data based on categories such as region, product, or time period. This enhances user interaction and provides customized views of the data for better analysis.



6.2 UML DIAGRAM

The UML diagram represents the interaction between the user and the Retail Sales Analytics System. It illustrates the key functionalities of the system and how the user interacts with different components to analyze retail data effectively.

In this system, the **user** is the primary actor who can perform multiple operations such as accessing the dashboard, applying filters, examining trends, and exporting insights. The **“View Dashboard”** function allows the user to visualize overall sales performance through interactive dashboards.

The **“Filter Data”** functionality enables users to refine data based on specific criteria such as product category, region, or time period. The **“Analyze Trends”** component helps in identifying patterns, growth trends, and performance metrics from the filtered data.

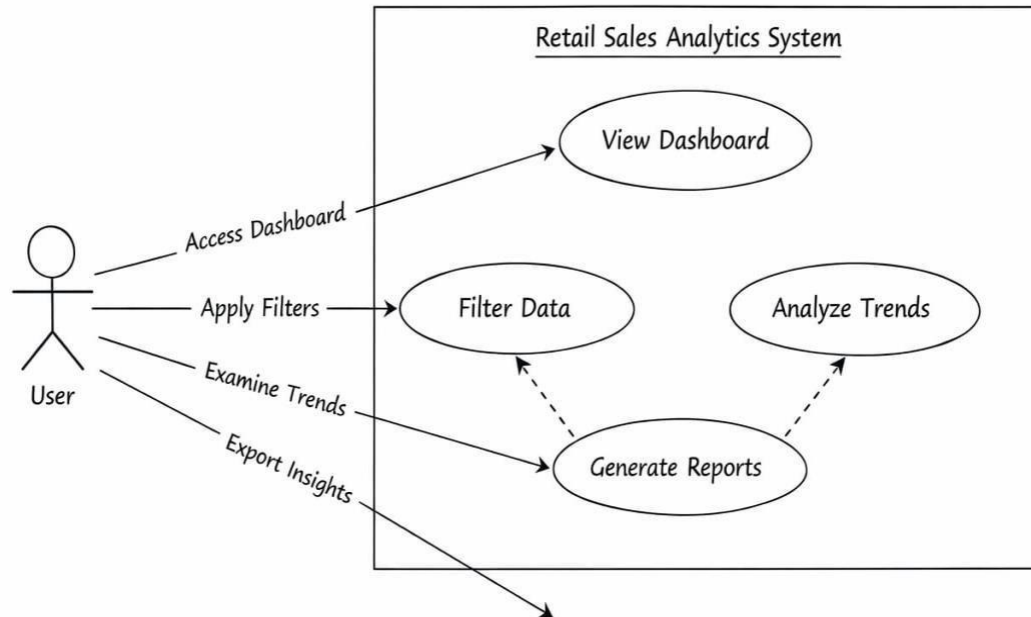
Additionally, the **“Generate Reports”** function compiles the analyzed data into structured reports, which can be used for business decision-making. These reports can also be exported for further use.

Overall, the UML diagram clearly demonstrates the system's workflow, highlighting how user actions trigger different processes to transform raw data into meaningful insights for effective retail analysis.

The diagram also highlights the relationship between different system functionalities. The **“Generate Reports”** component acts as a central process that depends on both filtered data and trend analysis. This indicates that accurate reporting is achieved only after proper data filtering and analytical processing. The dashed connections in the diagram represent dependencies between these processes, showing how different modules of the system are interconnected.

Furthermore, the UML diagram emphasizes the interactive and user-driven nature of the system. Users can dynamically apply filters and explore data in real time, which enhances flexibility and usability. This design ensures that even non-technical users can easily navigate the system and extract valuable insights. Overall, the system promotes efficient data analysis, better visualization, and informed decision-making in retail operations.

UML Diagram of Retail Sales Analytics System



CHAPTER 7 IMPLEMENTATION

7. KEY PERFORMANCE METRICS IMPLEMENTATION (USING DAX)

In this project, various Key Performance Indicators (KPIs) are implemented using Data Analysis Expressions (DAX) in Power BI. These measures help in analyzing sales performance, customer behavior, and operational efficiency. Each metric is calculated using specific formulas and visualized using dashboards for better decision-making.

1. Total Sales

- Total Sales represents the overall revenue generated from all transactions. It is calculated by summing the net amount of all orders in the dataset.
- DAX Formula:
- $\text{Total Sales} = \text{SUM}(\text{'Table' [Net Amount (₹)]})$
- This metric provides a high-level view of business performance and is displayed using KPI cards. For example, the total revenue generated is ₹12,50,000.

2. Total Orders

- Total Orders indicates the total number of transactions completed. It is calculated by counting the number of order IDs present in the dataset.
- DAX Formula:
- $\text{Total Orders} = \text{COUNT}(\text{'Table' [Order ID]})$
- This helps in understanding the volume of sales activity. For instance, the system recorded 2,450 total orders.

3. Average Order Value

- Average Order Value (AOV) measures the average revenue generated per order. It is calculated by dividing total sales by the total number of orders.
- DAX Formula:
- $\text{Avg Order Value} = \text{DIVIDE}([\text{Total Sales}], [\text{Total Orders}], 0)$
- This metric helps in evaluating customer purchasing behavior. In this case, the average order value is ₹510 per order.

4. Sales Growth Percentage

- Sales Growth Percentage shows the increase or decrease in sales compared to the previous time period. It is useful for analyzing business growth trends.
- DAX Formula:
- $\text{Sales Growth \%} = \text{DIVIDE}([\text{Total Sales}] - \text{CALCULATE}([\text{Total Sales}], \text{DATEADD}('Table'[\text{Order Date}], -1, \text{MONTH})), \text{CALCULATE}([\text{Total Sales}], \text{DATEADD}('Table'[\text{Order Date}], -1, \text{MONTH})), 0)$
- This metric indicates a positive growth of approximately 12%, showing improvement in sales performance.

5. Delivery Days

- Delivery Days calculates the time taken to deliver orders from the order date to the delivery date.
- DAX Formula:
- $\text{Delivery Days} = \text{DATEDIFF}('Table'[\text{Order Date}], 'Table'[\text{Delivery Date}], \text{DAY})$
- This helps in evaluating logistics efficiency. The average delivery time is between 3 to 5 days.

6. Cancellation Rate

- Cancellation Rate measures the percentage of orders that were cancelled. It is an important metric for assessing customer satisfaction and operational issues.
- DAX Formula:
- $\text{Cancellation Rate} = \text{DIVIDE}(\text{CALCULATE}(\text{COUNT}('Table'[\text{Order ID}]), 'Table'[\text{Order Status}] = "Cancelled"), [\text{Total Orders}], 0)$
- The system shows an 8% cancellation rate, indicating areas for improvement in service quality.

7. Running Sales

- Running Sales represents cumulative sales over time. It helps in tracking growth trends and understanding how revenue builds up over a period.

- DAX Formula:
- Running Sales =
CALCULATE([Total Sales],
FILTER(ALL('Table'[Order Date]),
'Table'[Order Date] <= MAX('Table'[Order Date])))
- This metric shows a gradual increase in sales (e.g., ₹2L → ₹5L → ₹9L), indicating consistent business growth.

1. TOTAL SALES

Code:

DAX

Total Sales =

SUM('Table'[Net Amount (₹)])

Result: ₹12,50,000 (Total revenue shown in Card)

2. TOTAL ORDERS

Code:

DAX

Total Orders =

COUNT('Table'[Order ID])

Result: 2,450 Orders

3. AVERAGE ORDER VALUE

Code:

DAX

Avg Order Value =

DIVIDE([Total Sales], [Total Orders], 0)

Result: ₹510 per order

4. SALES GROWTH %

Code:

DAX

Sales Growth % =

DIVIDE([Total Sales] -

CALCULATE([Total Sales], DATEADD('Table'[Order Date], -1, MONTH)),

CALCULATE([Total Sales], DATEADD('Table'[Order Date], -1, MONTH)),0)

Result: +12% growth

5. DELIVERY DAYS

Code:

DAX

Delivery Days =

DATEDIFF('Table'[Order Date],

'Table'[Delivery Date], DAY)

Result: 3–5 days per order

6. CANCELLATION RATE

Code:

DAX

Cancellation Rate =

```
DIVIDE(CALCULATE(COUNT('Table'[Order ID]),  
'Table'[Order Status]="Cancelled"),  
[Total Orders],0)
```

Result: 8% cancelled order

7. RUNNING SALES

Code:

DAX

Running Sales =

```
CALCULATE([Total Sales],  
FILTER(ALL('Table'[Order Date]),  
'Table'[Order Date] <= MAX('Table'[Order Date])))
```

Result: Gradual increase (e.g., 2L → 5L → 9L)

CHAPTER 8

RESULTS

8.1 OVERVIEW OF RESULTS

The implementation of the Retail Business Intelligence system successfully generated meaningful insights from the dataset. By using Power BI dashboards and DAX measures, various key performance indicators (KPIs) were analyzed to evaluate sales performance, customer behavior, and operational efficiency. The results are presented through interactive visualizations, enabling easy interpretation and decision-making.

8.2 SALES PERFORMANCE RESULTS

The analysis of total sales shows that the business generated a revenue of **₹12,50,000**, indicating strong overall performance. The sales growth percentage of **+12%** demonstrates a positive upward trend compared to the previous period. The running sales visualization further confirms consistent growth over time.

8.3 ORDER ANALYSIS RESULTS

The system recorded a total of **2,450 orders**, reflecting a high level of customer activity. The average order value (AOV) was calculated to be **₹510 per order**, which provides insight into customer spending behavior. These metrics help in understanding sales volume and revenue contribution.

8.4 DELIVERY AND OPERATIONS RESULTS

The average delivery time was found to be between **3 to 5 days**, indicating efficient logistics performance. However, the cancellation rate of **8%** highlights potential areas for improvement in order management and customer satisfaction.

8.5 VISUALIZATION RESULTS

The use of dashboards and charts such as line charts, bar charts, pie charts, and KPIs enabled clear visualization of data. Users were able to interact with slicers and filters to explore specific data segments, making the system highly user-friendly and effective for analysis.

CHAPTER 9 CONCLUSION

9.CONCLUSION

The project, *Comprehensive Retail Business Intelligence & Sales Performance Analysis, Optimization and Visualization of Data*, successfully demonstrates how data-driven approaches can enhance decision-making in the retail sector. By integrating data from multiple sources and applying analytical techniques, the system effectively transforms raw data into meaningful insights. The use of Power BI dashboards and DAX measures enables clear visualization of key performance indicators such as total sales, order volume, growth rate, and customer behavior.

The implementation of the system highlights the importance of real-time analysis and interactive reporting in modern retail environments. Through various visualizations and metrics, the project provides a comprehensive understanding of sales trends, operational efficiency, and customer patterns. These insights help businesses identify strengths, address weaknesses, and make informed strategic decisions to improve overall performance.

Furthermore, the project emphasizes the role of advanced analytics and visualization tools in simplifying complex datasets. By presenting information in an easy-to-understand format, the system ensures accessibility for both technical and non-technical users. This promotes a datadriven culture within organizations, encouraging the use of analytics for everyday decisionmaking.

In conclusion, the proposed Retail Business Intelligence system proves to be an effective solution for optimizing business operations, increasing profitability, and enhancing customer satisfaction. It provides a scalable and flexible framework that can be adapted to various retail environments. The project successfully meets its objectives and demonstrates the significant impact of Business Intelligence in achieving sustainable growth in the retail industry.

The project, *Comprehensive Retail Business Intelligence & Sales Performance Analysis, Optimization and Visualization of Data*, successfully demonstrates how data analytics and visualization can be used to improve decision-making in the retail sector. By integrating data from multiple sources and applying DAX-based analysis in Power BI, the system effectively converts raw data into meaningful insights. Key performance indicators such as total sales, order volume, growth rate, and customer behavior were analyzed, providing a clear understanding of business performance.

CHAPTER 10

FUTURE SCOPE

10.FUTURE SCOPE

The proposed Retail Business Intelligence system can be further enhanced by integrating advanced technologies and expanding its capabilities. One important area of improvement is the incorporation of real-time data streaming, which would allow businesses to monitor sales and operations instantly as transactions occur. Additionally, the system can be extended with advanced machine learning models to provide more accurate demand forecasting, customer behavior prediction, and personalized product recommendations.

Another potential enhancement is the deployment of the system on cloud platforms, enabling better scalability, accessibility, and data security. The integration of mobile applications can also allow users to access dashboards and insights anytime and anywhere. Furthermore, the system can be improved by incorporating artificial intelligence techniques such as automated decision support and chat-based analytics. These advancements will make the system more intelligent, efficient, and adaptable to the evolving needs of the retail industry, ensuring long-term growth and competitiveness.

In addition to the existing capabilities, the system can be enhanced by integrating advanced data integration techniques to combine structured and unstructured data sources such as social media, customer reviews, and online feedback. This would provide deeper insights into customer sentiment and preferences, enabling businesses to design more effective marketing strategies and improve customer engagement.

Another area of future development is the implementation of advanced security mechanisms to protect sensitive business and customer data. Features such as data encryption, role-based access control, and secure authentication can be incorporated to ensure data privacy and compliance with industry standards. This will increase user trust and ensure the safe handling of large volumes of retail data.

The system can also be extended to support multi-channel retail environments by integrating both offline and online sales platforms into a unified dashboard. This would allow businesses to track performance across different channels and gain a holistic view of their operations. Such integration will help in better resource allocation, improved customer experience, and more effective business strategies.

activity. This proactive approach enables quicker decision-making and helps businesses respond immediately to potential issues or opportunities.

CHAPTER 11 REFERENCES

11. REFERENCES

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DECLARATION

We hereby declare that the Real Time Research project report titled “Comprehensive Retail Business Intelligence & Sales Performance Analysis optimization and visualisation of the data”, submitted in partial fulfilment of the requirements for the award of the **Bachelor of Technology (B. Tech)** degree in **Computer Science & Engineering (Data Science)**, is a record of original work carried out by us under the supervision of **Mrs.shirisha**. This work has not been submitted for the award of any other degree or diploma at any other institute or university. In accordance with ethical practices in academic reporting, proper acknowledgments have been made wherever the work or findings of others have been cited.

				
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Guide

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